



Linux Handbook for DevOps, Cloud & Cybersecurity

This handbook provides a structured and practical introduction to Linux for students pursuing careers in DevOps Engineering, Cloud Computing, and Cybersecurity. It is designed as a foundational reference for both classroom learning and hands-on labs.

1. Introduction to Linux

Linux is an open-source operating system widely used in enterprise environments.

It powers cloud servers, DevOps pipelines, and cybersecurity platforms.

Understanding Linux is a mandatory skill for modern IT professionals.

2. Linux and Career Paths

DevOps: Linux hosts CI/CD tools, automation scripts, and containers.

Cloud Computing: AWS EC2, Azure VMs, and GCP instances run Linux.

Cybersecurity: Monitoring, logging, and system hardening occur on Linux.

3. Linux File System Explained

Linux uses a hierarchical file system starting at the root (/).

/home stores user files.

/etc stores system configuration.

/var stores logs and variable data.

/bin and **/usr/bin** contain essential commands.

/tmp stores temporary files.

4. Basic Navigation Commands

`pwd` – Display current directory.

`ls / ls -la` – List directory contents and permissions.

`cd` – Change directory.

`clear` – Clear terminal screen.

5. File & Directory Management

mkdir – Create directories.

touch – Create files.

cp – Copy files.

mv – Move or rename files.

rm – Delete files or directories.

6. Viewing & Editing Files

cat – Display file content.

less – Scroll through files.

head / tail – View file beginning or end.

nano / vim – Edit text files.

7. User & Process Management

whoami – Display current user.

id – Display user information.

sudo – Execute commands with elevated privileges.

ps – View running processes.

top – Monitor system resources.

8. Networking Commands

`ip a` – View network interfaces.

`ping` – Test network connectivity.

`ss / netstat` – Display open ports.

`curl / wget` – Test web connectivity.

9. Permissions & Security

Linux uses read, write, and execute permissions.

chmod – Change permissions.

chown – Change ownership.

Correct permissions prevent data breaches.

10. Linux in DevOps

Automation scripts run on Linux.

Containers and orchestration rely on Linux.

Configuration management tools manage Linux servers.

11. Linux in Cloud Computing

Cloud servers are Linux virtual machines.

Security groups and SSH protect access.

System updates and monitoring occur at OS level.

12. Linux in Cybersecurity

Security logs reveal suspicious activity.

Hardening reduces attack surfaces.

Incident response begins with Linux investigation.